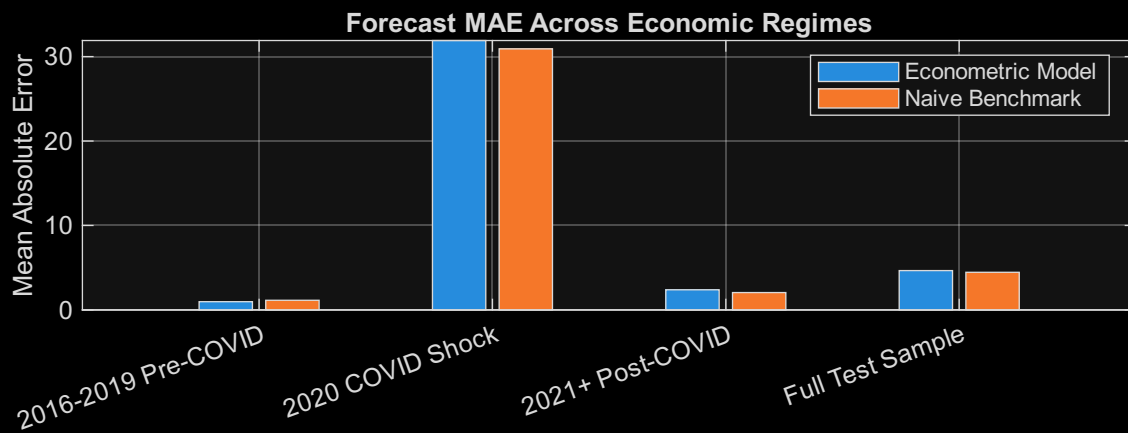
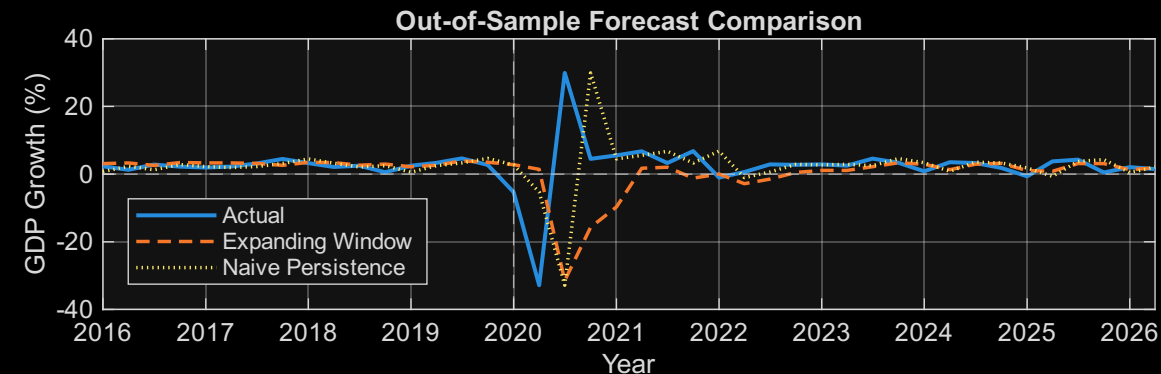
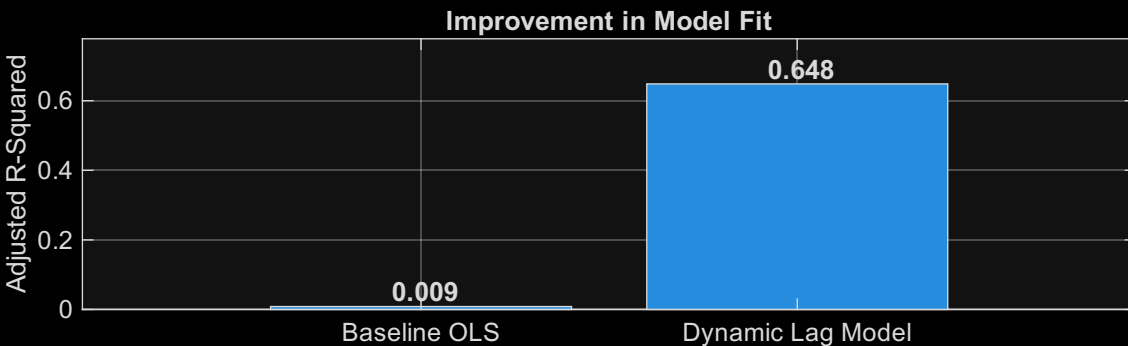
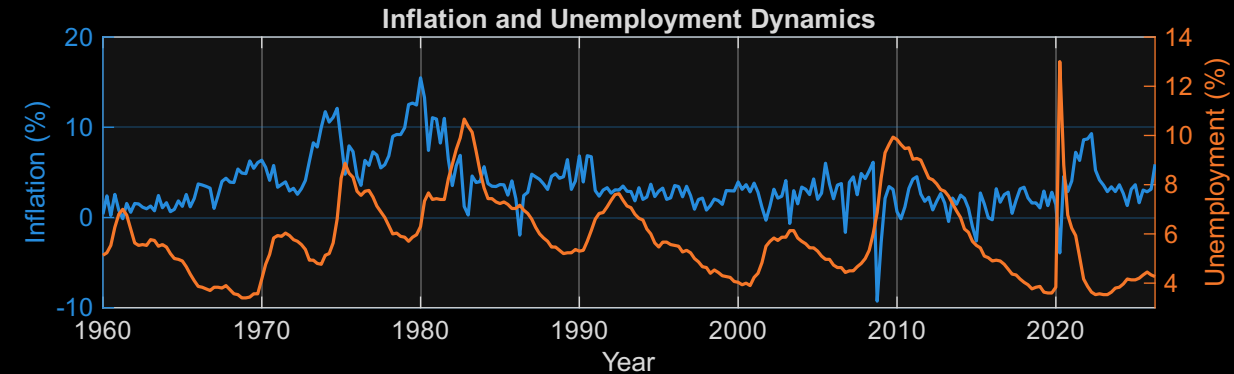
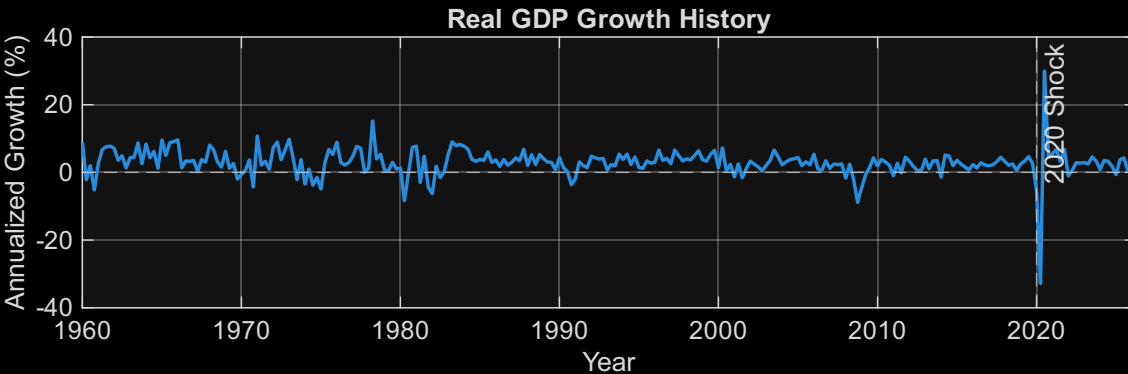


U.S. Macroeconomic Dynamics & GDP Growth Forecasting

MATLAB Econometrics Project | FRED Quarterly Data | 1960–2026



EXECUTIVE FINDINGS

Sample: 266 quarterly observations (1960–2026)

Baseline OLS Adjusted R^2 : 0.85%

Dynamic Model Adjusted R^2 : 64.83%

Pre-COVID RMSE improvement vs naive: 5.58%

Pre-COVID MAE improvement vs naive: 14.81%

2020 Structural Break F-statistic: 12.43

Structural Break p-value: 0.0000

Best full-sample OOS RMSE model: Naive Persistence

Best full-sample OOS RMSE: 11.53

Key conclusion:

Dynamic lag structures dramatically improve historical explanatory fit, but structural instability limits forecast generalization during major economic shocks.